

When AI Thinks With Us

An Exploratory Probe into AI-Assisted Learning and Cognitive Delegation

Tasneem Fatima | September 2026

Research Exploration 01 | Exploratory pilot study | $n = 5$

Abstract

Generative AI is increasingly becoming part of activities previously performed primarily by people, including explanation, summarisation, problem solving and feedback. This creates a human-centred design question: when AI assists with cognitive work, how do people decide what to retain as their own work and what to delegate? This exploratory study presents a small qualitative probe with five participants aged 19–25. Participants first responded to a short scenario about a university student deciding how to use AI while preparing for examinations. They then viewed a standardised AI response advocating an independent-first, AI-supported approach and answered the same question again without looking back at the response. A final question asked whether the AI response had changed their understanding of how the student should study.

The pilot is descriptive rather than causal. Across the five participants, responses repeatedly positioned AI as an aid for clarification, difficult concepts, summarisation, practice and feedback, while independent understanding, active recall and verification remained important. Three participants reported that the AI response changed their understanding somewhat, one reported very little change, and one reported no change. These self-reports do not measure learning or cognitive change. The findings instead suggest a direction for further research: the sequencing and boundary of AI assistance may matter, particularly whether people attempt a task independently before delegating part of it to AI.

1. Motivation and Research Context

My professional work has centred on enterprise technology, digital transformation, workflow improvement and, more recently, AI-enabled products and agents. That experience led to a question that cannot be answered through product delivery alone: what happens to the human participant in a workflow when AI moves from being a tool that is consulted to an active contributor to cognitive work?

Research on cognitive offloading provides a useful theoretical starting point. Risko and Gilbert (2016) describe cognitive offloading as the use of external action to alter the information-processing demands of a task and review both the mechanisms that prompt offloading and its cognitive consequences. Recent research has begun to examine related questions in generative-AI-mediated learning, including whether AI functions as a substitute for thinking or as a scaffold that preserves cognitive agency (Hashmi et al., 2026; Zhu et al., 2026).

This pilot does not attempt to establish whether AI improves or harms learning. Instead, it explores how people describe the boundary between independent cognitive work and AI assistance. The

distinction matters for human-centred design because the same AI capability can function as a scaffold, a source of feedback, or a substitute for thinking depending on how the interaction is structured.

2. Research Questions

- RQ1. How do participants describe appropriate uses of AI when a learner is deciding how much of their studying to do independently?
- RQ2. After exposure to a standardised AI recommendation, what changes, if any, appear in participants' descriptions of appropriate AI use?
- RQ3. What early patterns suggest how people may decide which parts of cognitive work to retain and which parts to delegate to AI?

3. Method

3.1 Participants

Five participants (P1-P5), aged 19-25, took part in the exploratory probe. Their backgrounds varied across science, technology, infrastructure, commerce and business/entrepreneurial activity. Participants are identified only by anonymous codes in this report.

3.2 Scenario and Procedure

Participants received a standardised scenario, 'The Overloaded Student'. Alex, a second-year university student, is preparing for examinations in four subjects while struggling with readings and assignments. Alex considers using AI to summarise difficult readings, explain concepts and generate practice questions, while also trying to preserve independent study and recall.

Participants first answered: 'Based only on the scenario, what do you think are the THREE most important things Alex should consider when deciding how to use AI for studying?' They were then shown the same standardised AI response:

“AI can help Alex summarize difficult material, explain concepts and generate practice questions. However, Alex should continue attempting to understand and recall material independently because actively retrieving information and explaining concepts can support learning. A balanced approach could involve attempting a problem independently first, then using AI for clarification, feedback or additional practice.”

The response was then hidden. Participants answered the question again without looking back at it and completed a self-report item asking whether the AI response had changed their understanding of how Alex should study. An optional final question asked which parts of studying they would personally do themselves and which they would delegate to AI.

3.3 Analytic Approach

The analysis is an exploratory thematic reading of the written responses. Its purpose was to identify recurring descriptions of AI use and boundaries around independent work, not to test statistical hypotheses. Observations are distinguished from interpretation and from hypotheses for future research. Substantive responses were retained even when participants did not provide exactly three items.

4. Findings

4.1 Self-reported change after exposure

Participant	Q3 response	Descriptive note
P1	D - Not at all	Independent-first view remained essentially unchanged.
P2	C - Very little	AI remained mainly a tool for summarising and clarifying difficult material.
P3	B - Yes, somewhat	Described trying independently before using AI for difficult concepts.
P4	B - Yes, somewhat	Described independent study followed by AI support for doubts and difficult topics.
P5	B - Yes, somewhat	Described a balanced model: independent study first, then AI for practice and feedback.

Three of five participants selected 'Yes, somewhat', one selected 'Very little', and one selected 'Not at all'. No participant selected 'Yes, substantially'. These figures describe perceived change only; with $n = 5$, they are not estimates of population-level effects.

4.2 Recurring themes

AI as augmentation rather than replacement. Participants frequently described AI as support for learning rather than a substitute for understanding, reading, solving or recalling.

Independent attempt before assistance. Several participants explicitly preferred trying to understand or solve material themselves before turning to AI.

Difficulty-contingent delegation. Participants often described AI as most useful when a concept was difficult, unclear or time-consuming.

Practice, recall and feedback. Practice questions, active recall, correction and mock tests appeared repeatedly in participants' descriptions of appropriate AI use.

Verification and critical judgment. Participants raised the possibility that AI outputs could be inaccurate and described checking or validating AI assistance.

Concern about dependence. Some responses explicitly mentioned avoiding dependency and retaining their own thinking or learning style.

4.3 Emerging interpretation

P1 already articulated an independent-first position before seeing the AI response and subsequently reported no change. P3 described a boundary between easier material handled independently and difficult material delegated to AI. P4 and P5 similarly described independent

study followed by AI support for clarification, practice or correction. These responses suggest that perceived task difficulty may be a useful variable for future research. This remains tentative: task difficulty was not objectively measured or manipulated, and stated preferences were not compared with observed behaviour.

5. Discussion

The most consistent observation in this small pilot is not that participants rejected AI, but that they placed conditions around its use. AI was welcomed for explanation, summarisation, practice and feedback, while participants continued to value independent understanding, active recall, verification and their own reasoning.

Cognitive offloading is a useful theoretical lens, but the present study should not be read as demonstrating cognitive offloading in a formal experimental sense. Participants described what they would or would not delegate to AI; the study did not measure cognitive effort, memory load, neural activity or subsequent learning. 'AI-supported cognitive delegation' is therefore used here as a descriptive phrase for participants' stated choices about which elements of a learning task they would retain and which they would ask AI to support.

The standardised AI response also creates an important methodological issue. Because it explicitly recommended independent-first, AI-supported learning, similarity in post-exposure answers may reflect priming, memory of the intervention, conformity, genuine reconsideration, or some combination of these. The pilot therefore raises a research question rather than answering one: does the sequencing of AI assistance shape how much cognitive work people retain for themselves?

6. Limitations

- Very small convenience sample (n = 5); findings are exploratory and not generalisable.
- Participants were young adults aged 19-25 and do not represent the full range of learners or knowledge workers.
- The scenario concerned studying rather than workplace collaboration; transfer to professional work is a research proposition, not a finding.
- The standardised AI response may have primed subsequent answers.
- Q3 measured perceived change, not actual changes in learning, retention, cognitive load or behaviour.
- No objective performance, delayed recall, transfer, confidence calibration or behavioural AI-use measures were collected.
- The analysis was researcher-interpreted and exploratory; no inter-rater reliability was calculated.
- P1 provided four substantive points in Q1 and two in Q2, while P4 provided more than three in Q2; these responses were retained rather than artificially reduced.
- One P4 statement ('less emphasis on accuracy') was ambiguous and was not used as a basis for interpretation.

7. Proposed Next-Stage Study

The pilot points toward a controlled study of how the timing and role of AI assistance affect human cognitive participation. A possible design would compare four interaction conditions:

- Human-first: independent attempt followed by AI feedback or clarification.
- AI-first: AI explanation or solution followed by human attempt.
- Human-AI collaboration: iterative alternation between independent reasoning and AI support.
- Human-only: no AI assistance.

Future measures could include task performance, immediate and delayed recall, transfer to novel problems, perceived cognitive load, confidence and calibration, amount and timing of AI assistance, and willingness to delegate different task components. Behavioural interaction logs could complement self-report. Depending on the research question and available infrastructure, NeuroIS methods could later be considered to investigate cognitive processes more directly; such methods are not required to interpret the present pilot.

8. From Learning to AI-Augmented Work

Although this probe uses a learning scenario, the underlying design problem extends to knowledge-intensive work. In both learning and professional settings, AI can explain, generate, summarise, recommend and solve. The human-centred question is where the boundary should sit: which parts of a task should remain with the person, when should AI intervene, and how can an AI system support performance without unintentionally weakening learning, judgment or agency?

A next step is to investigate how these questions change when AI-supported practices become embedded in collaborative knowledge work. In organisations, AI-assisted outputs are not only produced by individuals; they are reviewed, explained, transferred and relied on by others. This extends the inquiry from cognitive participation at the individual level toward human agency, shared knowledge and continuity in AI-augmented work. The present pilot does not provide evidence about workplace collaboration, but it helps specify questions that can be examined through future qualitative and controlled research.

9. Conclusion

This exploratory study began with a practical question: when AI can perform part of a cognitive task, what do people choose to keep for themselves? Across five participants, AI was generally described as useful for clarification, difficult concepts, summarisation, practice and feedback, while independent understanding, active recall, verification and personal reasoning remained important. Three participants reported a moderate change in their understanding after viewing the standardised AI response, but the study cannot determine whether this represented genuine conceptual change, priming or another response effect.

The value of the pilot is therefore not a claim about whether AI improves or harms learning. Its value is in identifying a concrete design and research problem: how should AI assistance be sequenced and bounded so that people can benefit from computational support while retaining meaningful cognitive participation? Answering that question rigorously requires controlled studies

that measure not only immediate performance, but also learning, transfer, cognitive effort, judgment and longer-term human capability.

References

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Appendix A. Study Materials

Scenario: The Overloaded Student

Alex is a second-year university student preparing for final examinations in four subjects. With two weeks remaining, Alex is struggling to keep up with readings and assignments and often studies late at night while moving between notes, online resources and social media. Alex decides to create a weekly study plan, divide material into smaller sections, practise with questions and take regular breaks. Alex is also considering using AI to summarise difficult readings, explain concepts and generate practice questions, but is unsure how much of the studying should be done independently and how much should be delegated to AI.

AI response shown to all participants

AI can help Alex summarize difficult material, explain concepts and generate practice questions. However, Alex should continue attempting to understand and recall material independently because actively retrieving information and explaining concepts can support learning. A balanced approach could involve attempting a problem independently first, then using AI for clarification, feedback or additional practice.

Questions

- Q1. Based only on the scenario, what do you think are the THREE most important things Alex should consider when deciding how to use AI for studying?
- Q2. Without looking back at the AI response, what are the THREE most important things you now think Alex should consider?
- Q3. Did the AI response change your understanding of how Alex should study? A. Yes, substantially. B. Yes, somewhat. C. Very little. D. Not at all. Briefly explain why.
- Optional. If you were Alex, which parts of your studying would you prefer to do yourself, and which would you delegate to AI? Why?